

TSCR SERIES

MODEL

TSCR-05E

The TSCR-05E collaborative robot brings automation to workspaces shared with people. Harmonic-drive joints, hand-guided teaching and safety-rated monitored stop let it run fence-free machine tending, assembly, testing and packaging cells. A compact 1052 mm build height and 138 × 138 mm base make it easy to place beside operators on existing benches.

6-Axis Payload 5 kg · Reach 800 mm	Repeatability ±0.03 mm positional accuracy	Texsonics RC In-house controller, drives and teach pendant	Site-ready Floor / Wall / Ceiling mount, factory calibrated
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01 SPECIFICATIONS

MECHANICAL

AXES	6
PAYLOAD	5 kg
MAX REACH	800 mm
REPEATABILITY	±0.03 mm
JOINTS	Harmonic drive, force-limited

ENVIRONMENT / CONTROL

HEIGHT	1052 mm
BASE SIZE	138 × 138 mm
TEACHING	Hand guiding + teach pendant
SAFETY	Collaborative operation per ISO/TS 15066 design intent
PROGRAMMING	Graphical HMI / Offline CAM

02 PRODUCT



TSCR-05E · TSCR SERIES

03 WORKING RANGE

MOTION — GROUP A	
J1 — BASE ROTATION	±170°
J2 — SHOULDER	±170°
J3 — ELBOW	±160°

MOTION — GROUP B	
J4 — WRIST 1	±170°
J5 — WRIST 2	±170°
J6 — TOOL ROTATION	±175°

04 OVERVIEW

WORKS WITH PEOPLE,
NOT BEHIND FENCES.

The TSCR-05E is a collaborative robot with force-limited harmonic-drive joints, hand-guided teaching and safety-rated monitored stop. Deployable on existing benches in days, not months.

01 **HAND-GUIDED TEACHING**
Move the arm through the motion and record waypoints — no programming language required.

02 **FORCE-LIMITED JOINTS**
Every joint monitors torque and halts on unexpected contact, per the design intent of ISO/TS 15066.

03 **SMALL FOOTPRINT**
A 800 mm reach in a 138 × 138 mm base fits beside operators on existing lines.

05 CONTROL & INTEGRATION

CONTROLLER
Texsonics RC series
in-house motion controller,
dual-kernel real-time OS

TEACH PENDANT
11.5" touch tablet
drag-teaching, Wi-Fi,
on-device offline
programming

FIELDBUS
**EtherCAT / Modbus
TCP**
industrial fieldbus,
dual-protocol support

SOFTWARE
TEXCAM offline CAM
toolpath generation, digital
twin simulation, vision

06 APPLICATIONS

Deployable across the primary applications the TSCR-05E was engineered for — full production-line integration with the Texsonics control stack.

Fence-free Machine Tending

Assembly

Testing & Inspection

Packaging

Machine Tending

CNC Loading

Palletizing

Vision Inspection

07

BUILT FOR THE OPERATOR

- Live 3D position display
- Online command execution
- Position editor
- Variable monitor
- I/O monitor
- Error diagnostics
- Dynamic teach-in
- Digital twin simulation

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SERVICE & SUPPORT**Made in Coimbatore.
Supported from
Coimbatore.**

Every TSCR-05E is engineered, built and calibrated in our Coimbatore facility. Field service, spares and applications engineering ship from the same address — no import lead times, no time-zone escalations.

STANDARD WARRANTY

12 months on the mechanical assembly, 12 months on the controller.

APPLICATIONS ENGINEERING

Cell design, cycle-time studies and end-of-arm tooling by our in-house team, included on every project.

SPARES & FIELD SERVICE

Direct-to-plant spares from Coimbatore, with pan-India field service engineers on 24-hour dispatch.

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READY TO INTEGRATE**Talk to an engineer about
integrating the TSCR-05E
into your line.**

Applications engineering, cycle-time sizing, and turnkey cell design are included on every project.

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TSCR-05E

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TSCR SERIES

6-AXIS • PAYLOAD 5 kg • REACH 800 mm

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